PROJECT WORK CS6811

SECOND REVIEW

Federated Learning for Code Generation with Hallucination-Aware Fine Tuning

TEAM DETAILS

Abhinavh P (2022503059),

Prabhakara Arjun R (20225035003),

Buvanes Srivardhan K (20225033037)

MENTOR

Dr. B. Thanasekhar,

Professor,

Computer Technology, MIT

DOMAIN: Large Language Models

INTRODUCTION

Large Language Models (LLMs) have remarkable capabilities in code generation ,enabling automated solutions to a wide range of software engineering tasks. However LLMs frequently suffer from hallucinations in code generation, producing syntactically valid but semantically incorrect programs, non-existent APIs, or logically flawed implementations. Existing approaches to mitigating code hallucinations primarily rely on prompt engineering, repeated sampling, or full model fine-tuning. Full fine-tuning is computationally expensive and impractical in low-resource or decentralized settings, and repeated sampling does not address the underlying causes and mitigation of hallucination. In real-world development environments, code generation systems are often deployed across multiple decentralized clients where data cannot be centrally aggregated due to privacy, ownership.

To address these limitations we propose a hallucination-aware code generation framework that combines program analysis, targeted automatic program repair, and federated parameter-efficient fine-tuning using LoRA

PROBLEM STATEMENT

Code generation using large language models (LLMs) is currently hindered by hallucinations. Existing mitigation strategies typically rely on supervised fine-tuning or reinforcement learning from human feedback (RLHF), where models are trained to align with human preferences .While parameter-efficient fine-tuning methods such as LoRA reduce computational overhead compared to full fine-tuning, both approaches primarily optimize model behavior but without explicit verification of program correctness. As a result, these methods may improve correctness but often fail to systematically detect and correct deeper structural, semantic, and execution-level errors in generated code.

Furthermore, in real world scenarios, high-quality code and execution feedback are distributed across multiple organizations making centralized data collection infeasible due to privacy concerns. Without a secure and decentralized learning mechanism, models are unable to effectively learn from diverse hallucination patterns observed across clients. This creates a critical gap for a unified framework that integrates explicit hallucination detection and automated program verification with privacy-preserving, parameter-efficient adaptation to improve the quality of code generation .

OBJECTIVES

- To design a hallucination-aware code generation framework that integrates static and dynamic program analysis.
- To detect and classify different types of hallucinations in generated code using oracle-based verification.
- To apply automatic program repair for correcting hallucinated code using knowledge-guided patching.
- To fine-tune large language models using parameter-efficient LoRA in a federated learning setting.
- To evaluate the proposed framework on standard code generation benchmarks and analyze hallucination reduction.

LITERATURE SURVEY

S.NO	TITLE	NAME OF THE JOURNAL	METHODOLOGY	LIMITATIONS
[1]	Advancing LLM-Generated Code Reliability: A Hybrid Approach for Hallucination Detection	IEEE Transactions on Software Engineering (TSE), 2025	• Proposes SDHD, a hybrid hallucination detection framework that integrates static analysis and dynamic execution for LLM-generated code. . • This framework detects eight hallucination types and achieves superior precision, recall, and F1-score across multiple benchmarks.	• Dynamic detection depends on test coverage quality, and incomplete coverage may miss certain logical failures. . • The framework is currently validated only on Python, limiting cross-language generalizability.
[2]	LLM-Based Test-Driven Interactive Code Generation: User Study and Empirical Evaluation	IEEE Transactions on Software Engineering (TSE), 2024	• TICODER is a test-driven interactive workflow that incrementally clarifies user intent through LLM-generated tests and lightweight user feedback. • The system ranks and prunes candidate code suggestions based on user-approved tests using two variants: PASS/FAIL validation and explicit output specification.	• There is interactive overhead, and incorrect or noisy user feedback can prune correct solutions prematurely. • Effectiveness is contingent on the LLM's ability to generate high-quality tests.

[3]	A Federated Adaptive Large Language Model Fine-Tuning Framework for Software Development	IEEE Transactions on Services Computing, 2025	• The framework integrates LoRA-based parameter-efficient fine-tuning within a federated setting, where each organization locally trains lightweight adapter modules and shares only adapter to the central server. • A FedAvg-style is used to combine adapters based on local data size and an L2-norm constraint to manage gradient divergence	• Experimental results show degraded performance on unrelated datasets. • Although communication costs are reduced, the framework still requires multiple communication rounds, which may introduce latency in large-scale federations.
[4]	Towards Efficient Fine-Tuning of Language Models With Organizational Data for Automated Software Review	IEEE Transactions on Software Engineering 2024	• Proposes CodeMentor, a framework for few-shot learning that utilizes heuristic rules and weak supervision to construct datasets from internal organizational data • It employs Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning and integrates Reinforcement Learning from Human Feedback (RLHF) to incorporate domain expertise.	• The framework depends on the quality of organizational data, which may introduce domain bias and limit generalization across different coding lang. • RLHF requires computational resources and expert involvement,bottleneck scenario during scaling.
[5]	The Impact of Prompt Programming on Function-Level Code Generatio	IEEE Transactions on Software Engineering (TSE), 2025	• Empirical study to evaluate the impact of prompt programming techniques on function-level code generation. • CodePromptEval, a dataset of 7,072 prompts derived from 221 real-world Python functions, covering five prompt techniques: few-shot learning, chain-of-thought, persona, function signature, and package context, including all 32 possible combinations.	• The study is restricted to Python function-level tasks, limiting generalizability to other languages and higher-level program synthesis. • Rely on benchmark-driven evaluation, which may not fully capture real-world scenario. • Prompt techniques are evaluated in a fixed ordering, excluding effects of alternative prompt sequencing.

[6]	An Adaptive Framework Embedded With LLM for Knowledge Graph Construction	IEEE Transactions on Multimedia, Vol. 27, 2025	• KG construction happens in three stages: Information Extraction (IE) for open triple extraction, Additional Relation (AR) for enriching triples with semantic information, and Knowledge Graph Normalization (KGN) for schema-level alignment using vector similarity search. • Adaptive Prompt Generation (APG) module automatically generates, evaluates, and ranks prompts to guide LLM behavior across stages, enabling multi-domain KG construction from text, speech, and images.	• High computational overhead due to multi-stage LLM inference, embedding generation, and vector similarity matching. • The framework relies heavily on prompt engineering, which,, is not fully autonomous and may still suffer from hallucination and semantic inconsistency.
[7]	Fight Fire With Fire: How Much Can We Trust ChatGPT on Source Code-Related Tasks?	IEEE Transactions on Software Engineering (TSE), Vol. 50, No. 12, 2024	• Evaluating self-verification capability of GPT for: code generation, code completion, and program repair. • Direct question, guiding question, and test report generation prompting is used assess whether ChatGPT can reliably validate its own outputs. • Effectiveness is measured using accuracy, precision, recall, and F1-score, with additional analysis of self-contradictory hallucinations.	• Results show that ChatGPT frequently overestimates the correctness of its own outputs, leading to low recall in direct self-verification. • While guiding questions and test reports improve recall, precision is reduced. • Generated test reports often contain incorrect explanations, especially for code generation and failed repairs. .

[8]	FedALoRA: Adaptive Local LoRA Aggregation for Personalized Federated Learning in LLM	IEEE Internet of Things Journal, Vol. 12, No. 24, December 2025	• Proposes FedALoRA, a personalized federated learning framework that integrates LoRA-based PEFT with adaptive local aggregation to address severe non-IID data in federated LLM training. • FedALoRA uses an Adaptive LoRA Aggregation to merge global and local parameters via locally learned weights. This selective, element-wise fusion focused on higher transformer layers preserves domain expertise while capturing shared knowledge without increasing communication costs.	• FedALoRA introduces additional local optimization complexity due to adaptive weight learning in ALoRA modules. • The method requires careful tuning of the hyperparameter *l*, which controls the number of transformer layers involved in adaptive aggregation, and improper settings can degrade performance.
[9]	No Need to Lift a Finger Anymore? Assessing the Quality of Code Generation by ChatGPT	IEEE Transactions on Software Engineering (TSE), Vol. 50, No. 6, June 2024	• Evaluation framework for assessing ChatGPT-based code generation. The authors construct standardized prompts for 728 LeetCode algorithm problems and 54 CWE security scenarios across five programming languages. • Generated code is evaluated using automated judgment platforms (LeetCode for functional correctness) and static analysis tools (CodeQL for vulnerability detection). • A multi-round fixing process is designed, where execution or analysis feedback is iteratively fed back	• ChatGPT performs significantly better on problems published before 2021, indicating reliance on memorized patterns rather than reasoning. • The dialog-based fixing process has limited effectiveness, for logically complex problems . • Generated code often contains security vulnerabilities and non-deterministic behavior.

[10]	Knowledge Enhanced Program Repair for Data Science Code	Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (ICSE), 2025	• API KG Construction is done using a custom ontology and RDF triples to model API attributes and dependencies for seven data science libraries. • SPARQL is used for knowledge retrieval from on buggy code. • AST level Bug Knowledge Enrichment, which executes code node-by-node to localize runtime and assertion errors at the AST node level.	• DS-KG is version-specific and requires continuous updates to track evolving API documentation, making long-term maintenance costly and error-prone. • KG-based knowledge retrieval introduces higher latency than plain-text search
[11]	THINK: Tackling API Hallucinations in LLMs via Injecting Knowledge	IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2025	• THINK proposes a two-phase API hallucination mitigation framework consisting of Pre and Post execution Optimization. • Constructs an API knowledge database (Javadoc + validated code examples) and retrieves task-relevant APIs using task decomposition, similarity search, and LLM-based API labeling to guide code generation. • Error Classification into Seven API Error Types for correction	• It may introduce irrelevant API knowledge, occasionally degrading performance for tasks already solvable by strong LLMs. • Both retrieval results and ReAct-based repair depend on LLM generation, leading to non-deterministic behavior and reduced reproducibility across runs

[12]	EvoAPR: Enhancing Large Language Models for Automatic Program Repair with Genetic Algorithm and Dynamic LoRA	IEEE International Conference on Web Services (ICWS), 2025	• Genetic evolution of buggy hunks under tri-objective fitness: syntax compliance, bug-hunk masking rate, repair-location masking rate. • Dynamic LoRA swaps non-overlapping rank-r adapters per bug-index/project for on-demand, project-specific parameter retrieval. • Base + LoRA models generate candidate patches; token-level entropy scorer picks the least uncertain fix as the final patch.	• Genetic template evolution and dynamic LoRA fine-tuning introduce significant computation and implementation complexity compared to direct prompting-based APR methods. • The effectiveness of both stages relies on accurate bug localization and rich bug context.
[13]	CodeHalu: Investigating Code Hallucinations in LLMs via Execution Based Verification	Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025	• Executes LLM-generated code against test cases under time and memory constraints, recording execution failures, unexpected outputs, infinite loop behaviors to detect hallucination states. • Aggregates execution states across multiple LLMs and applies a two-stage heuristic induction process to classify hallucinations into four main types (Mapping, Naming, Resource, Logic) and eight subcategories.	• The study focuses exclusively on Python, and hallucination behaviors in other programming languages are not evaluated. • CodeHalu targets functional and logical correctness; identifying or mitigating security vulnerabilities is outside its scope.

ARCHITECTURE DIAGRAM OF THE PROPOSED SYSTEM

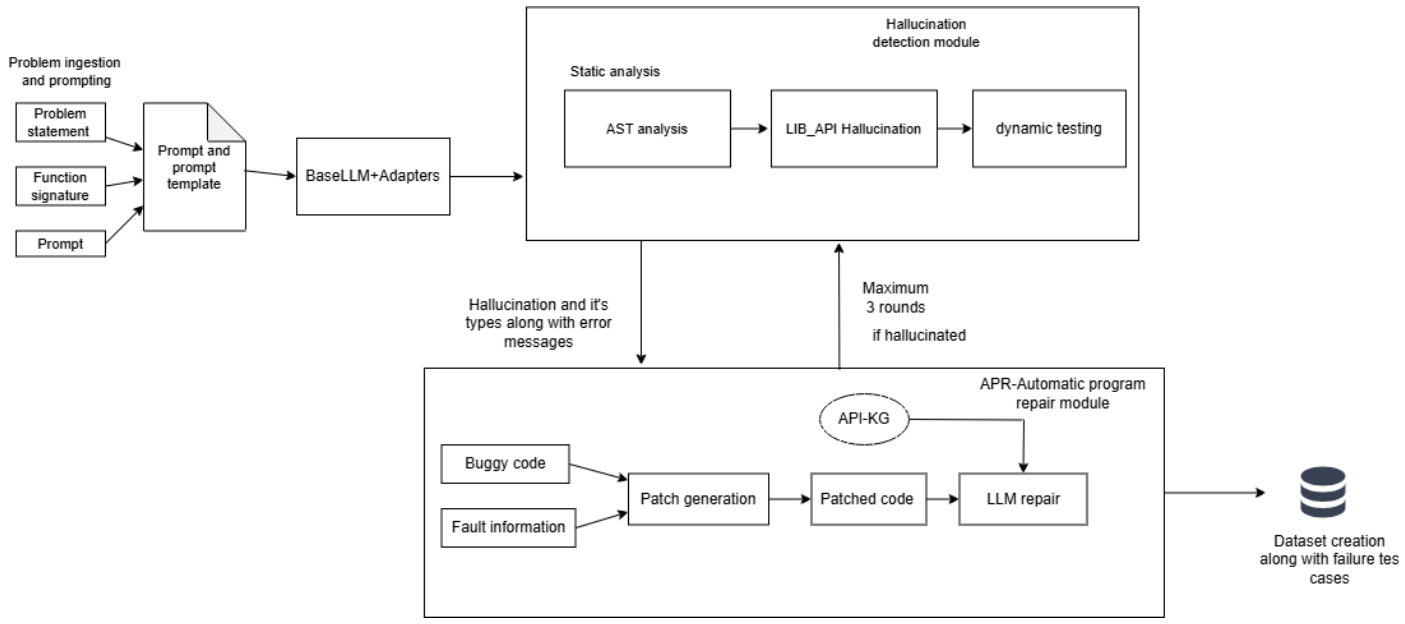


Fig1. Client side hallucination detection and mitigation framework

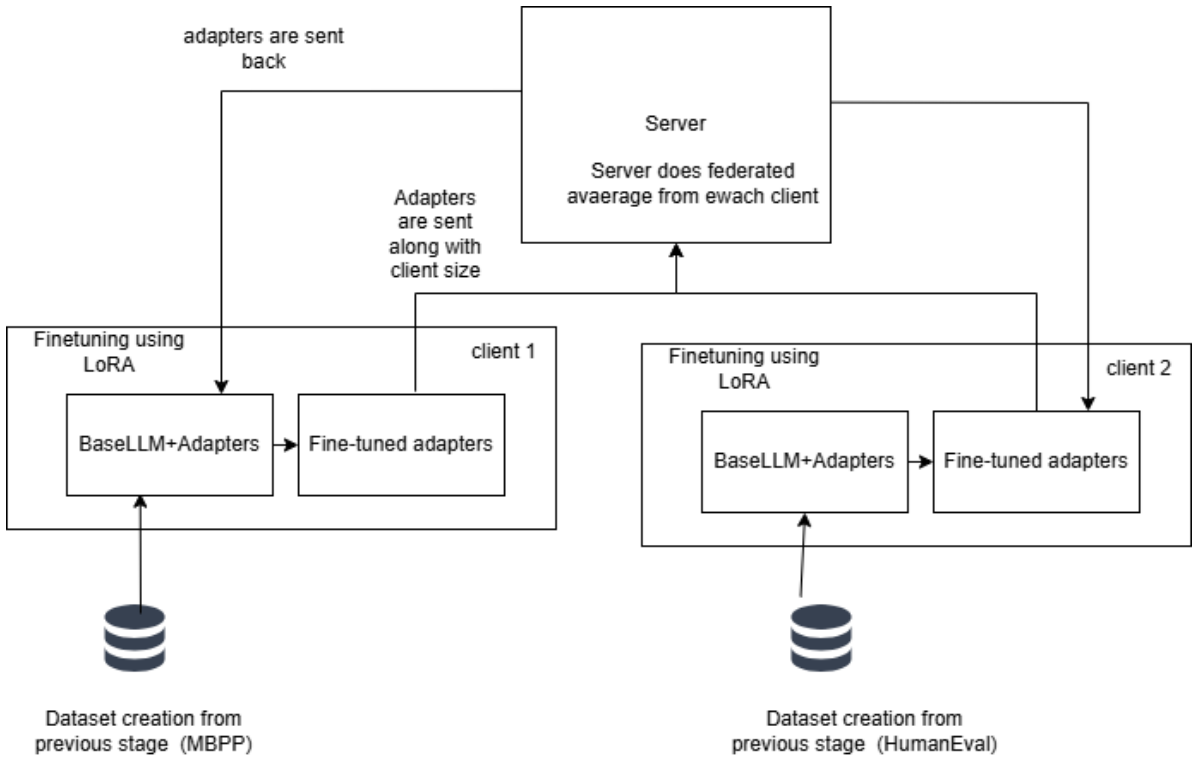


Fig 2. Federated learning + LoRA for fine-tuning

IDENTIFICATION OF PROJECT MODULES, TOOLS & PLATFORMS

S.NO	MODULES	SUB - MODULES	TOOLS & PLATFORMS
1.	**Problem ingestion and code generation**	• Dataset selection . • Client-wise dataset partitioning . • Single-shot prompting. • Pretrained LLM selection and loading. • Baseline inference using frozen LLM	Hugging Face Datasets, Transformers, Python, Pandas, NumPy, Google Colab.
2.	**Base Model Selection and Configuration**	• Static analysis using AST ,Library analysis. • Structured hallucination labeling.	Python ast module
3.	**Automatic Program Repair (APR)**	• Knowledge extraction. • Patch-based repair • Bounded repair attempts. • Ground-truth verification and hard hallucination labeling.	Numpy ,pandas, sklearn , scipy, statsmodel , Qwen2.5-Coder (Repair), Python.
4.	**Fine tuning using LoRA**	• Hallucination-aware dataset construction • Low-rank decomposition (r = 8). • Injection of LoRA modules into Query and Key projections. • Parameter isolation with frozen backbone. • Adapter checkpointing.	PyTorch, Hugging Face PEFT, Transformers.

5.	**Federated Aggregation and Distribution**	• Secure client-side adapters • Federated Averaging (FedAvg) with sample-size-based weighting. • Global adapter construction	Flower ,PyTorch, NumPy
6.	**Evaluation and analysis**	• Evaluation on MBPP, HumanEval, and DS-1000 benchmarks. • Functional correctness using Pass@1. • Hallucination detection metrics (Precision, Recall, F1). • Code similarity analysis using CodeBLEU. • Cross-client validation. • Before–after performance comparison.	CodeBLEU, Matplotlib, Seaborn, Scikit-learn

METHODS / MODELS

1. Problem ingestion and code generation

Input:

Benchmark code-generation datasets.

Output:

Initial LLM-generated code solutions (potentially containing hallucinations).

Methods:

- **Dataset Loading:** Benchmark code-generation datasets including MBPP, HumanEval, and DS-1000 are loaded using the Hugging Face library.
- **Client-wise Dataset Allocation:** Each federated client is assigned a distinct dataset or dataset subset to simulate heterogeneous behaviour.
- **Problem Specification Construction:** For each task, structured inputs consisting of the problem statement, function signature, constraints is constructed.
- **Single-Shot Prompting:** Code generation is performed using a single-shot prompting strategy, where the model receives the task specification without iterative refinement.
- **Baseline Inference:** The frozen Qwen2.5-Coder-3B pretrained model is used to generate initial code outputs, which may contain hallucinations and serve as baseline for downstream analysis.

2. Hallucination detection

Input:

LLM-generated code obtained from the code generation module.

Output:

Detected hallucination types along with structured diagnostic information.

Methods:

- **AST Construction and Analysis**: Abstract Syntax Tree (AST) is constructed and analyzed to detect syntax errors, undefined variables, and invalid language constructs**.**
- **Library analysis:** We use ast to check if each methods,functions align with the library in static setup.
- **Dynamic Analysis:** The test cases are executed on the code to detect runtime errors and semantic hallucinations.
- **Hallucination Labeling:** Detected errors are categorized into different hallucination types, and the resulting information is recorded for dataset construction during fine-tuning.

3. Automatic program repair module

Input:

Buggy code, detected hallucination type, fault hints, and a domain knowledge graph suggestions.

Output:

Corrected code validated against ground-truth oracles or hard hallucination label.

Methods:

- **Knowledge Extraction:** library, class, method, and dependency information is extracted from dir,help,inspect .
- **Patch-Based Repair:** Applying localized patches to the hallucinated code from fault information from previous modules**.**
- **Fault-Guided Prompting:** The detected hallucination type, fault hints, and relevant knowledge graph entries are provided to LLM for code correction**.**
- **Bounded Repair Attempts :** program repair is performed with a bounded number of attempts to ensure controlled behavior.
- **GT Verification:** Each candidate is then verified with Hallucination detection module. If verification fails after bounded attempts,it is marked as hard hallucination.

4. Fine tuning using LoRA

Input:

Hallucination-aware code samples, including buggy code, diagnostic information, and repair outcomes, along with a frozen pre-trained LLM.

Output:

Client-specific LoRA adapters trained on hallucination aware datasets.

Methods:

- **Dataset Construction:** Construct structured dataset using problem description, buggy code, hallucination type, repairability flag and corrected code.
- **Low-Rank Decomposition:** Weight updates are parameterized using two low-rank matrices A and B, where the rank r is set to 8.
- **Attention Layer Injection:** LoRA modules are injected into the Query and Key projection matrices of each Transformer layer.
- **Parameter Isolation:** Original model weights remain frozen, and only LoRA parameters are made trainable.
- **Adapter Checkpointing:**Only trained LoRA adapter weights are saved and transmitted to the server, ensuring lightweight communication and privacy preservation.

5. Federated Aggregation and Distribution

Input:

Client-side LoRA adapter weights, along with sample size.

Output:

Global LoRA adapter distributed to all participating clients.

Methods:

- **Adapter Collection:** LoRA adapter weights are collected from all participating clients after local fine-tuning.
- **Federated Averaging:** Client-side adapters are aggregated using the Federated Averaging (FedAvg) algorithm, where clients with larger sample sizes contribute proportionally higher weights to the global adapter.
- **Global Adapter Construction:** A global LoRA adapter is constructed by averaging the weighted client adapters while keeping the base model parameters frozen.

- **Adapter Distribution:** The aggregated global LoRA adapter is redistributed to all clients and combined with the shared base model for subsequent inference or further training rounds.

6. Evaluation and analysis

Input:

Code generated by the base model and the federated LoRA-enhanced model across benchmark datasets.

Output:

Benchmark-wise evaluation.

Methods:

- **Functional Correctness (Pass@1):**The Pass@1 metric is used to measure functional correctness if code passes all unit tests on the first attempt.
- **Hallucination Detection Metrics:** Precision, Recall, and F1-score are computed based on oracle-validated hallucination labels to assess the accuracy of hallucination detection.
- **Code Similarity Analysis:** CodeBLEU is used as a secondary metric to measure syntactic and structural similarity between generated code and reference implementations.
- **Cross-Client Validation:** Cross-validation is performed across federated clients to evaluate generalization and robustness under heterogeneous data distributions.
- **Before–After Comparison:**Model performance is compared before and after LoRA fine-tuning to quantify improvements in correctness and hallucination reduction.

IMPLEMENTATIONRESULTS

MODULE 1

Figure 3,4,5 shows the data loading for three dataset HumanEval, MBPP and DS1000 respectively.

Figure 6 shows the code snippet for code generation with QwenCoder template, which involves tokenization, decoding and code extraction.

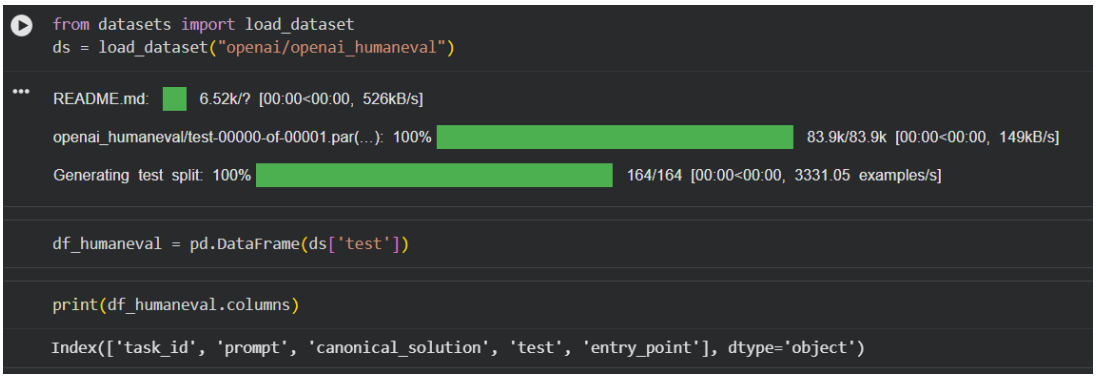


Fig 3

Fig 4

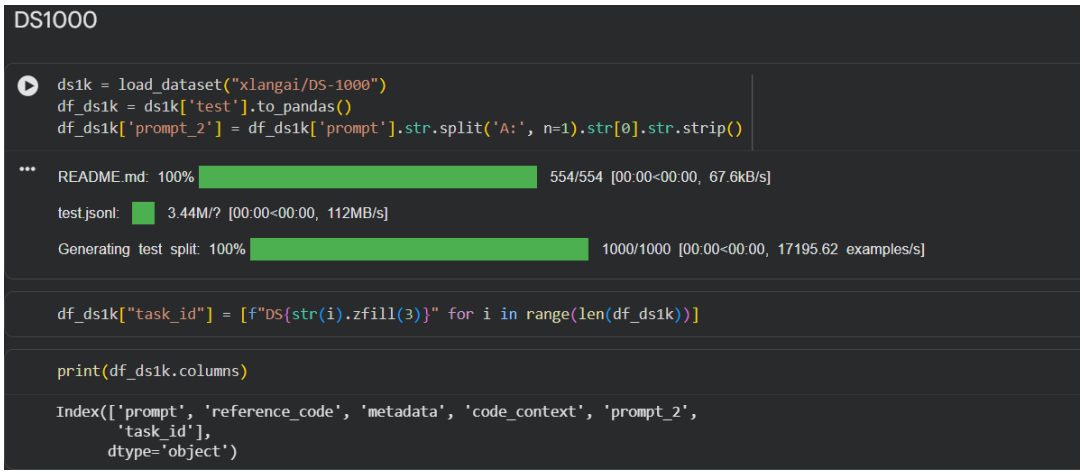


Fig 5

```
formatted_messages = construct_prompt(row['prompt'])
inputs = tokenizer.apply_chat_template(
    formatted_messages,
    add_generation_prompt=True,
    tokenize=True,
    return_dict=True,
    return_tensors="pt"
).to("cuda")

with torch.no_grad():
    outputs = model.generate(
        **inputs,
        max_new_tokens=512,
        do_sample=False,
        pad_token_id=tokenizer.eos_token_id
    )

gen_ids = outputs[0][len(inputs['input_ids'][0]):]
raw_response = tokenizer.decode(gen_ids, skip_special_tokens=True)
clean_code = extract_python_code(raw_response)
```

Fig 6

MODULE 2

Figure 7 shows hallucination distribution across all 3 datasets classified into syntax, indentation errors, structural hallucinations.

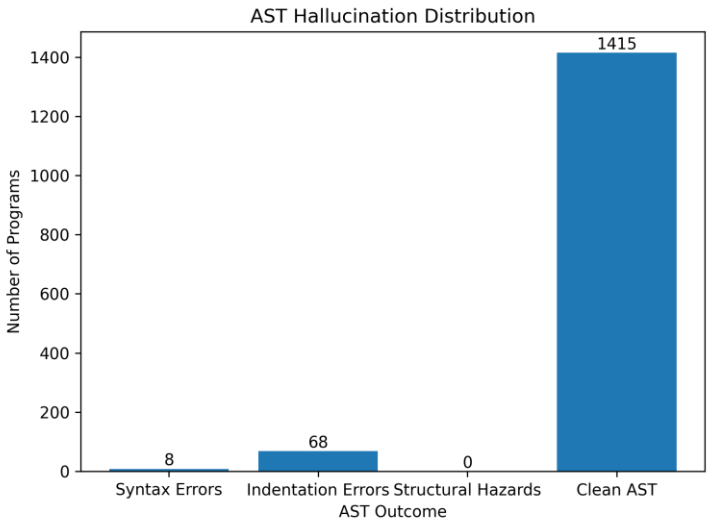


Fig 7

Figure 8 shows library or api hallucination across all dataset.

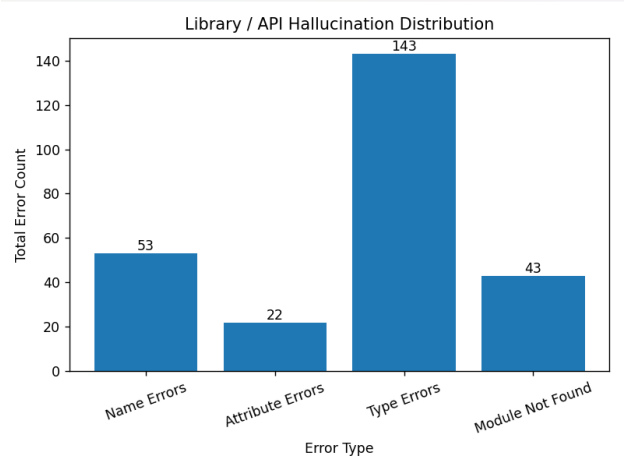


Fig 8

Figure 9 & 10 shows results after dynamic hallucination. Fig 9 describes the different type of errors encountered in a dataset. Fig 10 gives the overall hallucination pass and fail after dynamic analysis.

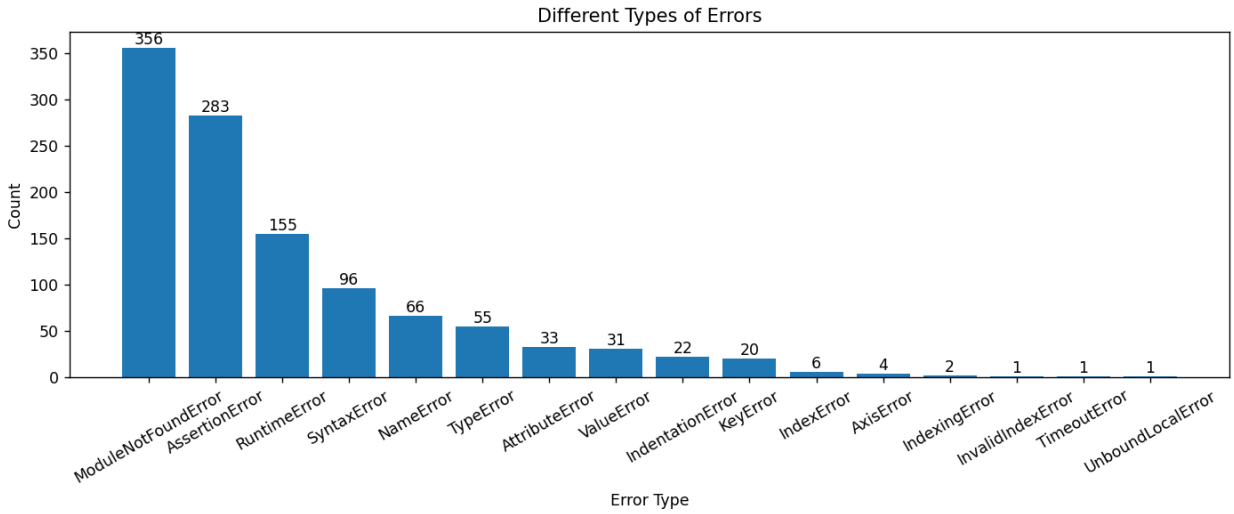


Fig 9

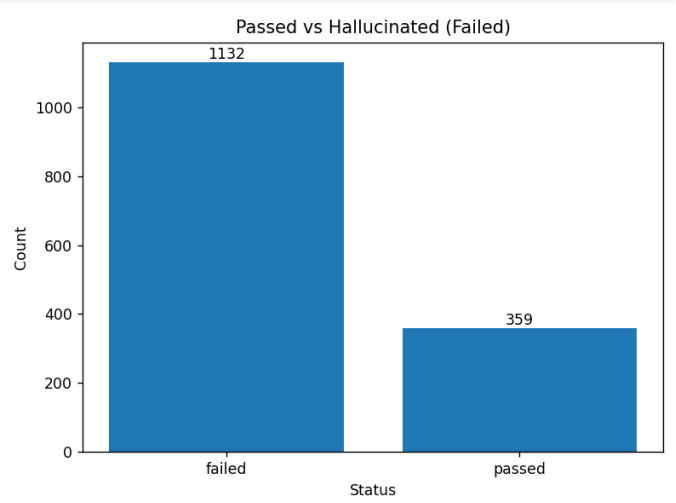


Fig 10

MODULE 3

Figure 11,12,13 shows an example from DS100 with task_id 42. We could see the patch generated around the error line and also the information to fix the error from fault_information.

```
df = pd.read_csv('test_ds1000.csv')
code = """
import pandas as pd
import numpy as np
df = test_input
result = df.iloc[1:].reset_index(drop=True).add_prefix('A_')
result = result.append(df.iloc[0].rename(columns=lambda x: f'B_{x}'))
"""
dataset_type = "ds1000"
task_id = "DS042"
row = task_row = df[df["task_id"] == "DS042"].iloc[0].to_dict()
result = do_pipeline(code,dataset_type,row,task_id)
print('Q RESULTS:')
print(result)
```

Fig 11

```
import pandas as pd
import numpy as np
df = test_input
result = df.iloc[1:].reset_index(drop=True).add_prefix('A_')
<<<< [ERROR START] (dynamic: AttributeError)
result = result.append(df.iloc[0].rename(columns=lambda x: f'B_{x}'))
[ERROR FINISH] (dynamic: AttributeError) >>>>
```

Fig 12

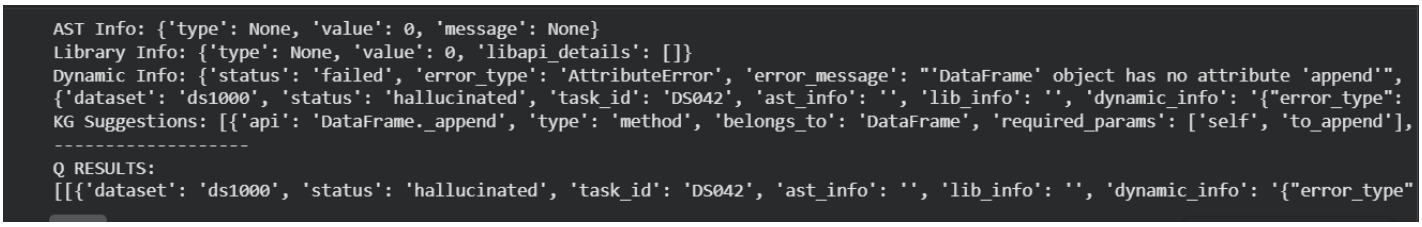


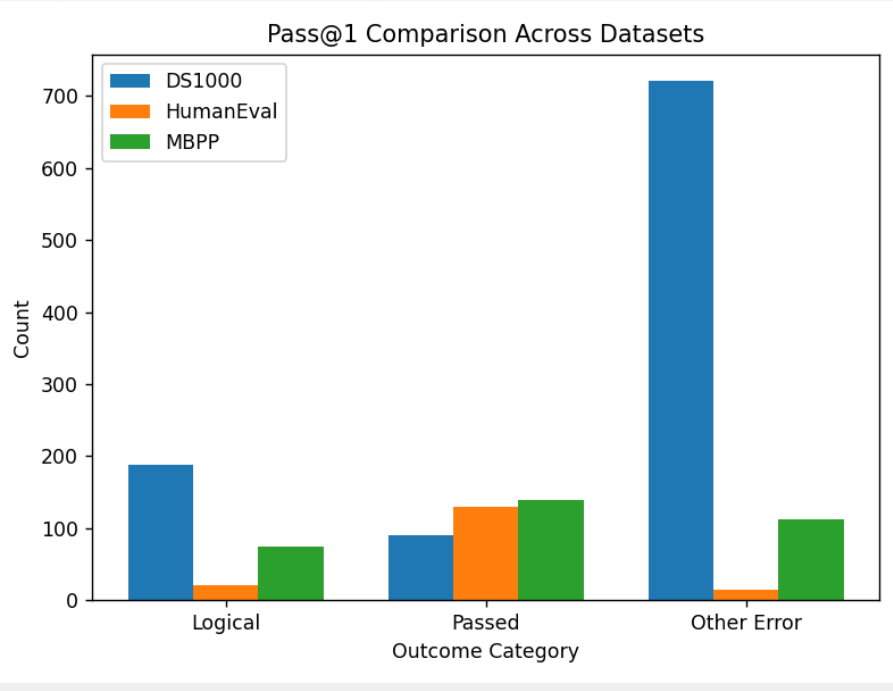
Figure 13

EXPERIMENTAL SETUP

- **Base LLM** :
    - QwenCoder2.5-3B parameter is used as base model.
    - Type: Causal Language Models
    - Training Stage: Pretraining & Post-training
    - Number of Parameters: 3.09B
    - Number of Paramaters (Non-Embedding): 2.77B
    - Number of Layers: 36
    - Number of Attention Heads (GQA): 16 for Q and 2 for KV
    - Context Length: Full 32,768 tokens
    - 4-bit Quantized Model Loading:The model was loaded using 4-bit quantization with the bitsandbytes library.
    - Deterministic Generation Settings: We configured do_sample = False and temperature = 1.0 to ensure deterministic and reproducible code generation across multiple runs.
- **Dataset**:
    - **HumanEval**: Consist of 164 rows and 5 features.
    - **MBPP**: Consist of 377 rows and 6 features. It is made up of most basic python programming.
    - **DS1000**: Consist of 1000 data science problems in python.
- **Development environment:**
    - **Google collab**: It is used for LLM inference, code generation ,building overall pipeline.It used NVIDIA T4 GPU with CUDA acceleration on runtime.
    - **VS Code**: Local system is used to build static and dynamic modules which doesn't require GPU computation. Building modules and analysis is done on local from the generated dataset.

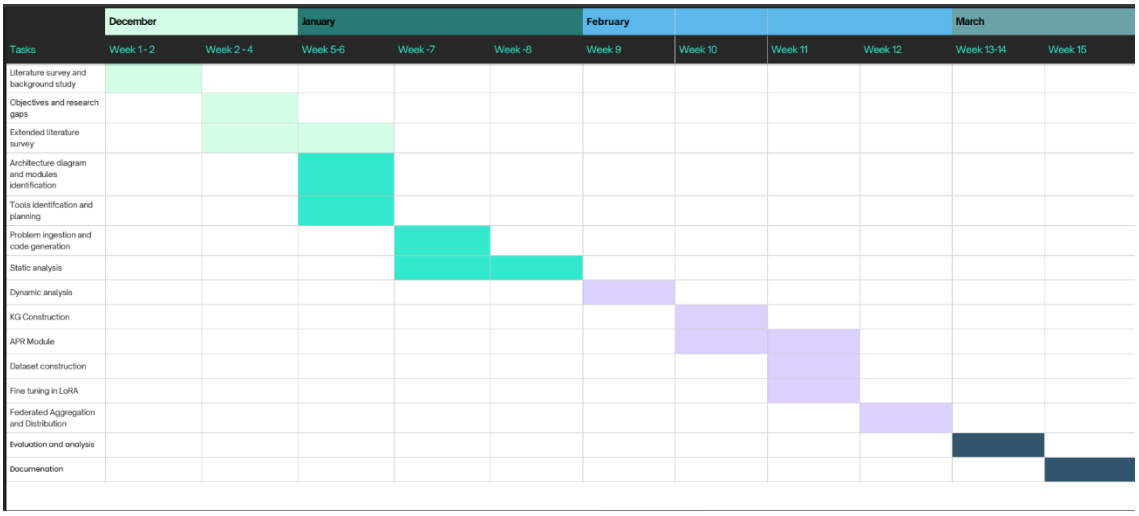
EVALUATION METRIC

Pass@1: The Pass@1 metric is used to measure functional correctness if code passes all unit tests on the first attempt.



Pass@1 comparison across dataset.

TIMELINE FOR COMPLETION OF MODULES



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